

Assignment 2- Data Analysis outputs

SECTION 1- DATA EXPLORATION

Q1. View first 10 records

The screenshot shows the MySQL Workbench interface with the 'superstore_analysis' database selected. The SQL editor contains the following queries:

```
1 CREATE DATABASE superstore_analysis;
2 USE superstore_analysis;
3 USE superstore_analysis;
4
5 SHOW TABLES;
6
7 -- SECTION 1 DATA EXPLORATION
8 /* Q1. View first 10 records */
9 SELECT * FROM superstore LIMIT 10;
```

The 'Result Grid' displays the first 10 rows of the 'superstore' table. The columns include Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, and Sub-Category.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42400	South	PUR-60-10001798	Furniture	Bookcases
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42400	South	PUR-CH-10000454	Furniture	Chairs
3	CA-2016-136688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-IA-10000240	Office Supplies	Labels
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Port Lauderdale	Florida	33311	South	PUR-TA-10000577	Furniture	Tables
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Port Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	PUR-FU-10001467	Furniture	Furnishings
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002853	Office Supplies	Art
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BS-100003910	Office Supplies	Binders
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances

Q2. Total number of records

The screenshot shows the MySQL Workbench interface with the 'superstore_analysis' database selected. The SQL editor contains the following queries:

```
4
5 SHOW TABLES;
6
7 -- SECTION 1 DATA EXPLORATION
8 /* Q1. View first 10 records */
9 SELECT * FROM superstore LIMIT 10;
10
11 /* Q2. Total number of records */
12
13 SELECT COUNT(*) AS Total_Rows FROM superstore;
```

The 'Result Grid' displays the result of the COUNT(*) query, showing a single row with the value 9694.

Total_Rows
9694

Q3. Table structure

MySQL Workbench interface showing the execution of SQL queries for Q3. The queries are:

```
9 • SELECT * FROM superstore LIMIT 10;
10
11 /* Q2. Total number of records */
12
13 • SELECT COUNT(*) AS Total_Rows FROM superstore;
14
15 /* Q3. Table structure */
16
17 • DESC superstore;
18
```

The results show the table structure of superstore:

Field	Type	Null	Key	Default	Extra
Row ID	int	YES			
Order ID	text	YES			
Order Date	text	YES			
Ship Date	text	YES			
Ship Mode	text	YES			
Customer ID	text	YES			
Customer Name	text	YES			
Segment	text	YES			
Country	text	YES			
City	text	YES			
State	text	YES			
Postal Code	int	YES			

The Action Output shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
16	23:42:58	SELECT * FROM superstore LIMIT 10	10 row(s) returned	0.000 sec / 0.000 sec
17	23:43:51	SELECT COUNT(*) AS Total_Rows FROM superstore LIMIT 0, 1000	1 row(s) returned	0.046 sec / 0.000 sec
18	23:44:47	DESC superstore	21 row(s) returned	0.031 sec / 0.000 sec

Q4. Distinct Categories

MySQL Workbench interface showing the execution of SQL queries for Q4. The queries are:

```
14
15 /* Q3. Table structure */
16
17 • DESC superstore;
18
19 /* Q4. Distinct Categories */
20
21 • SELECT DISTINCT Category FROM superstore;
22
23
```

The results show the distinct categories:

Category
Furniture
Office Supplies
Technology

The Action Output shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
17	23:43:51	SELECT COUNT(*) AS Total_Rows FROM superstore LIMIT 0, 1000	1 row(s) returned	0.046 sec / 0.000 sec
18	23:44:47	DESC superstore	21 row(s) returned	0.031 sec / 0.000 sec
19	23:45:51	SELECT DISTINCT Category FROM superstore LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec

SECTION 2-FILTERING

Q5. Orders from West Region

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Assignment2solutions_Q1_Q27 SQL File 3

Limit to 1000 rows

```
21 SELECT DISTINCT Category FROM superstore;
22 -- SECTION 2 FILTERING
23 /* Q5. Orders from West Region */
24
25 SELECT * FROM superstore WHERE Region = 'West';
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DH-13845	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	DM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage

superstore 19 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	23:44:47	DESC superstore	21 row(s) returned	0.031 sec / 0.000 sec
19	23:45:51	SELECT DISTINCT Category FROM superstore LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec
20	23:47:19	SELECT * FROM superstore WHERE Region = 'West' LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec

Q6. Technology Products

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Assignment2solutions_Q1_Q27 SQL File 3

Limit to 1000 rows

```
21 SELECT DISTINCT Category FROM superstore;
22 -- SECTION 2 FILTERING
23 /* Q5. Orders from West Region */
24
25 SELECT * FROM superstore WHERE Region = 'West';
26
27 /* Q6. Technology Products */
28
29 SELECT * FROM superstore WHERE Category = 'Technology';
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones
20	CA-2014-143336	8/27/2014	9/11/2014	Second Class	ZD-21925	Zuschuss Donatell	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories
36	CA-2016-117990	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SH-20730	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	TEC-PH-10000486	Technology	Phones
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazemias	Corporate	United States	Naper-ville	Illinois	60540	Central	TEC-PH-10004093	Technology	Phones
46	CA-2016-118255	3/11/2016	3/13/2016	First Class	OH-18715	Odella Nelson	Corporate	United States	Eagan	Minnesota	55122	Central	TEC-AC-10000171	Technology	Accessories
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-AC-10002167	Technology	Accessories
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-PH-10003988	Technology	Phones

superstore 20 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	23:45:51	SELECT DISTINCT Category FROM superstore LIMIT 0, 1000	3 row(s) returned	0.015 sec / 0.000 sec
20	23:47:19	SELECT * FROM superstore WHERE Region = 'West' LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec
21	23:48:08	SELECT * FROM superstore WHERE Category = 'Technology' LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.016 sec

Q7. Sales greater than 1000

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Assignment2solutions_Q1_Q27 SQL File 3

Limit to 1000 rows

```
24
25 SELECT * FROM superstore WHERE Region = 'West';
26 /* Q6. Technology Products */
27
28 SELECT * FROM superstore WHERE Category = 'Technology';
29
30 /* Q7. Sales greater than 1000 */
31
32 SELECT * FROM superstore WHERE Sales > 1000;
33
```

Result Grid

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emly Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14405	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East	TEC-PH-10002447	Technology	Phones
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sived	Corporate	United States	Gilbert	Arizona	85234	West	OFF-AR-10002671	Office Supplies	Art
150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central	FUR-CH-10000454	Furniture	Chairs
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	TEC-MA-10000822	Technology	Machines
168	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	FUR-CH-10004287	Furniture	Chairs
216	CA-2015-146262	1/2/2015	1/9/2015	Standard Class	VW-21775	Victoria Wilson	Corporate	United States	Medina	Ohio	44256	East	TEC-MA-10000864	Technology	Machines

superstore 21 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	23:47:19	SELECT * FROM superstore WHERE Region = 'West' LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec
21	23:48:08	SELECT * FROM superstore WHERE Category = 'Technology' LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.016 sec
22	23:48:57	SELECT * FROM superstore WHERE Sales > 1000 LIMIT 0, 1000	467 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Q8. Furniture orders from East region

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Assignment2solutions_Q1_Q27 SQL File 3

Limit to 1000 rows

```
27
28 SELECT * FROM superstore WHERE Category = 'Technology';
29
30 /* Q7. Sales greater than 1000 */
31
32 SELECT * FROM superstore WHERE Sales > 1000;
33 /* Q8. Furniture orders from East region */
34
35 SELECT * FROM superstore WHERE Category='Furniture' AND Region='East';
36
```

Result Grid

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases
30	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East	FUR-CH-10003968	Furniture	Chairs
97	CA-2017-161018	11/9/2017	11/11/2017	Second Class	PH-18775	Parhena Norris	Home Office	United States	New York City	New York	10009	East	FUR-FU-10000629	Furniture	Furnishings
111	CA-2017-146780	12/25/2017	12/30/2017	Standard Class	CV-12805	Cynthia Voltz	Corporate	United States	New York City	New York	10035	East	FUR-FU-10001934	Furniture	Furnishings
120	CA-2016-103730	6/12/2016	6/15/2016	First Class	SC-20725	Steven Cartwright	Consumer	United States	Wilmington	Delaware	19805	East	FUR-FU-10002157	Furniture	Furnishings
141	CA-2016-103366	9/5/2016	9/7/2016	Second Class	JD-15895	Jonathan Doherty	Corporate	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings
147	CA-2014-100072	10/22/2014	10/28/2014	Standard Class	MG-17680	Maureen Gastineau	Home Office	United States	Newark	Ohio	43055	East	FUR-FU-10000521	Furniture	Furnishings
178	US-2015-101511	11/21/2015	11/23/2015	Second Class	JE-15745	Joel Eaton	Consumer	United States	Newark	Ohio	43055	East	FUR-CH-10004698	Furniture	Chairs

superstore 22 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	23:48:08	SELECT * FROM superstore WHERE Category = 'Technology' LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.016 sec
22	23:48:57	SELECT * FROM superstore WHERE Sales > 1000 LIMIT 0, 1000	467 row(s) returned	0.016 sec / 0.000 sec
23	23:49:43	SELECT * FROM superstore WHERE Category='Furniture' AND Region='East' LIMIT 0, 1000	584 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SECTION 3- AGGREGATION

Q9. Total Sales

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Assignment2solutions_Q1_Q27 SQL File 3*

Limit to 1000 rows

```
32 • SELECT * FROM superstore WHERE Sales > 1000;
33 /* Q8. Furniture orders from East region */
34
35 • SELECT * FROM superstore WHERE Category='Furniture' AND Region='East';
36
37 -- SECTION 3 AGGREGATION
38 /* Q9. Total Sales */
39
40 • SELECT SUM(Sales) AS Total_Sales FROM superstore;
41
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Total_Sales
2272449.8562999545

Schema: superstore_analysis

Result 23 x

Output

#	Time	Action	Message	Duration / Fetch
22	23:48:57	SELECT * FROM superstore WHERE Sales > 1000 LIMIT 0, 1000	467 row(s) returned	0.016 sec / 0.000 sec
23	23:49:43	SELECT * FROM superstore WHERE Category='Furniture' AND Region='East' LIMIT 0, 1000	584 row(s) returned	0.000 sec / 0.000 sec
24	23:50:44	SELECT SUM(Sales) AS Total_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

Q10. Average Sales

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Assignment2solutions_Q1_Q27 SQL File 3*

Limit to 1000 rows

```
34
35 • SELECT * FROM superstore WHERE Category='Furniture' AND Region='East';
36
37 -- SECTION 3 AGGREGATION
38 /* Q9. Total Sales */
39
40 • SELECT SUM(Sales) AS Total_Sales FROM superstore;
41 /* Q10. Average Sales */
42
43 • SELECT AVG(Sales) AS Average_Sales FROM superstore;
44
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Average_Sales
234.41818199917006

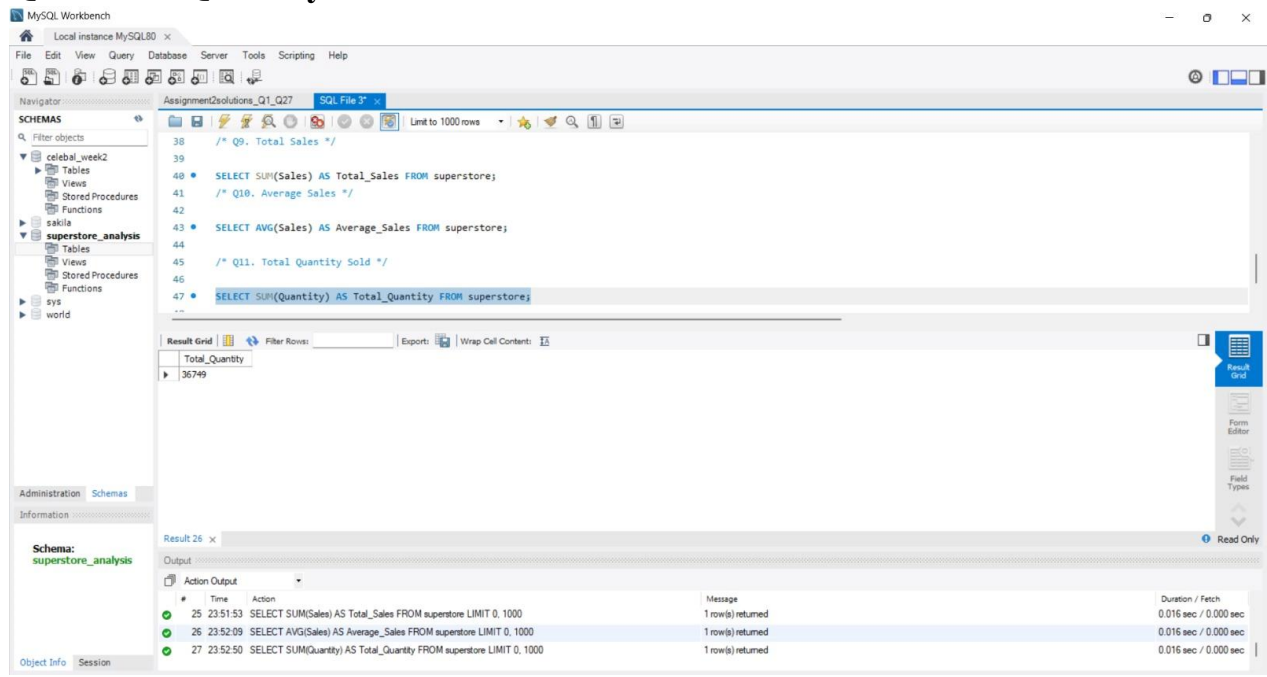
Schema: superstore_analysis

Result 25 x

Output

#	Time	Action	Message	Duration / Fetch
24	23:50:44	SELECT SUM(Sales) AS Total_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
25	23:51:53	SELECT SUM(Sales) AS Total_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
26	23:52:09	SELECT AVG(Sales) AS Average_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

Q11. Total Quantity Sold



MySQL Workbench interface showing the execution of a query to calculate the total quantity sold.

SQL Query:

```
/* Q9. Total Sales */
38
39
40 SELECT SUM(Sales) AS Total_Sales FROM superstore;
41
/* Q10. Average Sales */
42
43 SELECT AVG(Sales) AS Average_Sales FROM superstore;
44
/* Q11. Total Quantity Sold */
45
46 SELECT SUM(Quantity) AS Total_Quantity FROM superstore;
47
--
```

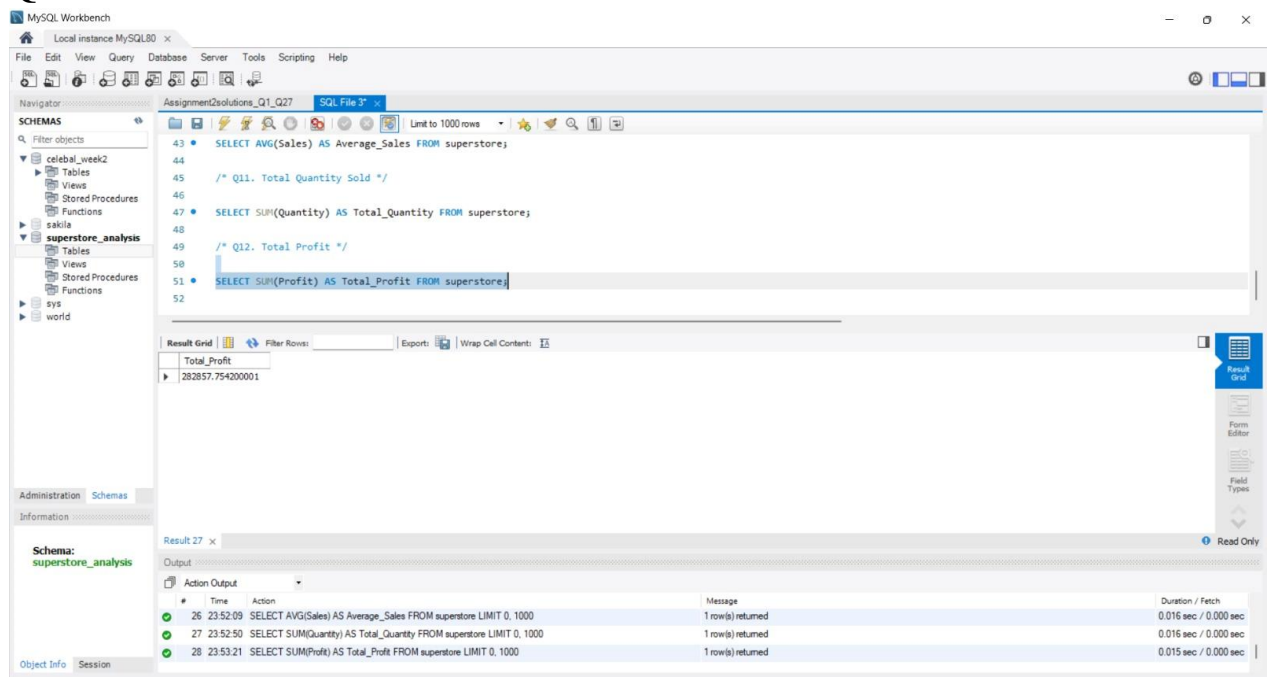
Result Grid:

Total_Quantity
36749

Action Output:

#	Time	Action	Message	Duration / Fech
25	23:51:53	SELECT SUM(Sales) AS Total_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
26	23:52:09	SELECT AVG(Sales) AS Average_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
27	23:52:50	SELECT SUM(Quantity) AS Total_Quantity FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec

Q12. Total Profit



MySQL Workbench interface showing the execution of a query to calculate the total profit.

SQL Query:

```
43 SELECT AVG(Sales) AS Average_Sales FROM superstore;
44
/* Q11. Total Quantity Sold */
45
46 SELECT SUM(Quantity) AS Total_Quantity FROM superstore;
47
/* Q12. Total Profit */
48
49
50 SELECT SUM(Profit) AS Total_Profit FROM superstore;
51
52
```

Result Grid:

Total_Profit
262857.754200001

Action Output:

#	Time	Action	Message	Duration / Fech
26	23:52:09	SELECT AVG(Sales) AS Average_Sales FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
27	23:52:50	SELECT SUM(Quantity) AS Total_Quantity FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
28	23:53:21	SELECT SUM(Profit) AS Total_Profit FROM superstore LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec

SECTION 4 - CATEGORY ANALYSIS

Q13. Sales by Category

The screenshot shows the MySQL Workbench interface with a SQL script for Q13. The script includes a comment for Q12 and a query for Q13. The results grid shows the total sales for each category.

```
49 /* Q12. Total Profit */
50
51 SELECT SUM(Profit) AS Total_Profit FROM superstore;
52 -- SECTION 4 CATEGORY ANALYSIS
53
54 /* Q13. Sales by Category */
55
56 SELECT Category, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Category
57 ORDER BY Total_Sales DESC;
58
```

Category	Total_Sales
Technology	835900.0669999964
Furniture	733046.8612999996
Office Supplies	703502.9280000031

Result 28 x

#	Time	Action	Message	Duration / Fetch
27	23:52:50	SELECT SUM(Quantity) AS Total_Quantity FROM superstore LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
28	23:53:21	SELECT SUM(Profit) AS Total_Profit FROM superstore LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
29	23:54:17	SELECT Category, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Category ORDER BY Total_Sales ...	3 row(s) returned	0.031 sec / 0.000 sec

Q14. Profit by Category

The screenshot shows the MySQL Workbench interface with a SQL script for Q14. The script includes a comment for Q13 and a query for Q14. The results grid shows the total profit for each category.

```
54 /* Q13. Sales by Category */
55
56 SELECT Category, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Category
57 ORDER BY Total_Sales DESC;
58 /* Q14. Profit by Category */
59
60 SELECT Category, SUM(Profit) AS Total_Profit FROM superstore
61 GROUP BY Category
62 ORDER BY Total_Profit DESC;
63
```

Category	Total_Profit
Technology	145387.0965999999
Office Supplies	120489.8864000001
Furniture	16980.7711999999

Result 29 x

#	Time	Action	Message	Duration / Fetch
28	23:53:21	SELECT SUM(Profit) AS Total_Profit FROM superstore LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
29	23:54:17	SELECT Category, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Category ORDER BY Total_Sales ...	3 row(s) returned	0.031 sec / 0.000 sec
30	23:55:57	SELECT Category, SUM(Profit) AS Total_Profit FROM superstore GROUP BY Category ORDER BY Total_Profit D...	3 row(s) returned	0.031 sec / 0.000 sec

SECTION 5- TOP PRODUCTS

The screenshot shows the MySQL Workbench interface with a SQL query in the editor and its results in the grid.

```
59
60 SELECT Category, SUM(Profit) AS Total_Profit FROM superstore
61 GROUP BY Category
62 ORDER BY Total_Profit DESC;
63
64 -- SECTION 5 TOP PRODUCTS
65 SELECT "Product Name", SUM(Sales) AS Total_Sales FROM superstore
66 GROUP BY "Product Name"
67 ORDER BY Total_Sales DESC
68 LIMIT 10;
```

Product Name	Total_Sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384
Cisco TelePresence System EX90 Videoconferen...	22638.48
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19021.479000000003
GBC Bomster 500 Manual ProClick Binding System	19024.5
Hewlett Packard LaserJet 3330 Copier	18539.686
HP DesignJet T520 Inkjet Large Format Printer ...	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.311999999998

Result 30 x

Output

#	Time	Action	Message	Duration / Fetch
1	00:13:17	SELECT "Product Name", SUM(Sales) AS Total_Sales FROM superstore GROUP BY "Product Name" ORDER BY ...	10 row(s) returned	0.031 sec / 0.000 sec

SECTION 6- TOP CONSUMERS

The screenshot shows the MySQL Workbench interface with a SQL query in the editor and its results in the grid.

```
66 GROUP BY "Product Name"
67 ORDER BY Total_Sales DESC
68 LIMIT 10;
69 -- SECTION 6 TOP CONSUMERS
70 /* Q16. Top 10 Customers by Sales */
71
72 SELECT "Customer Name", SUM(Sales) AS Total_Sales FROM superstore
73 GROUP BY "Customer Name"
74 ORDER BY Total_Sales DESC
75 LIMIT 10;
```

Customer Name	Total_Sales
Sean Miller	25043.05
Tamara Chand	19017.847999999998
Raymond Buch	15117.339
Tom Ashbrook	14595.62
Adrian Barton	14355.610999999997
Sanjit Chand	14142.339999999999
Ken Lonsdale	14071.917
Hunter Lopez	12873.297999999999
Sanjit Engle	12209.438000000002
Christopher Conant	12129.072

Result 31 x

Output

#	Time	Action	Message	Duration / Fetch
1	00:13:17	SELECT "Product Name", SUM(Sales) AS Total_Sales FROM superstore GROUP BY "Product Name" ORDER BY ...	10 row(s) returned	0.031 sec / 0.000 sec
2	00:14:35	SELECT "Customer Name", SUM(Sales) AS Total_Sales FROM superstore GROUP BY "Customer Name" ORDER B ...	10 row(s) returned	0.031 sec / 0.000 sec

SECTION 7- REGION ANALYSIS

Q17. Sales by Region

The screenshot shows the MySQL Workbench interface with a SQL query executed. The query is as follows:

```
74 ORDER BY Total_Sales DESC
75 LIMIT 10;
76 -- SECTION 7 REGION ANALYSIS
77 /* Q17. Sales by Region */
78
79 SELECT Region, SUM(Sales) AS Total_Sales FROM superstore
80 GROUP BY Region
81 ORDER BY Total_Sales DESC;
82
83 -- SECTION 8 BUSINESS USE CASE
```

The result grid shows the following data:

Region	Total_Sales
West	713471.3445000004
East	672194.0539999981
Central	497800.8728000007
South	389983.58500000037

The Action Output pane shows the execution details for three queries:

#	Time	Action	Message	Duration / Fetch
1	00:13:17	SELECT 'Product Name', SUM(Sales) AS Total_Sales FROM superstore GROUP BY 'Product Name' ORDER BY ...	10 row(s) returned	0.031 sec / 0.000 sec
2	00:14:35	SELECT 'Customer Name', SUM(Sales) AS Total_Sales FROM superstore GROUP BY 'Customer Name' ORDER B...	10 row(s) returned	0.031 sec / 0.000 sec
3	00:16:12	SELECT Region, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Region ORDER BY Total_Sales DES...	4 row(s) returned	0.015 sec / 0.000 sec

SECTION 8- BUSINESS USE CASE

Q18. Monthly Sales Trend

The screenshot shows the MySQL Workbench interface with a SQL query executed. The query is as follows:

```
83 -- SECTION 8 BUSINESS USE CASE
84 /* Q18. Monthly Sales Trend */
85
86 SELECT
87 MONTH(STR_TO_DATE('Order Date', '%m/%d/%Y')) AS Month_No,
88 SUM(Sales) AS Total_Sales
89 FROM superstore
90 GROUP BY Month_No
91 ORDER BY Month_No;
92
```

The result grid shows the following data:

Month_No	Total_Sales
1	94539.34159999999
2	59012.82540000001
3	203719.26379999987
4	135387.35799999996
5	153513.30969999999
6	151039.43330000006
7	145623.85000000002
8	157642.25099999984
9	303536.66570000002
10	198440.00270000002
11	348834.55700000001
12	321160.99850000003

The Action Output pane shows the execution details for four queries:

#	Time	Action	Message	Duration / Fetch
2	00:14:35	SELECT 'Customer Name', SUM(Sales) AS Total_Sales FROM superstore GROUP BY 'Customer Name' ORDER ...	10 row(s) returned	0.031 sec / 0.000 sec
3	00:16:12	SELECT Region, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Region ORDER BY Total_Sales DE...	4 row(s) returned	0.015 sec / 0.000 sec
4	00:18:29	SELECT MONTHSTR_TO_DATE('Order Date', '%m/%d/%Y') AS Month_No, SUM(Sales) AS Total_Sales FRO...	12 row(s) returned	0.015 sec / 0.000 sec

Q19. Monthly Profit Trend

MySQL Workbench

Local instance MySQL80

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Navigator

Assignment2solutions_Q1_Q27 SQL File 3

Limit to 1000 rows

```
92
93 /* Q19. Monthly Profit Trend */
94
95 SELECT
96     MONTH(STR_TO_DATE('Order Date', '%m/%d/%Y')) AS Month_No,
97     SUM(Profit) AS Total_Profit
98 FROM superstore
99 GROUP BY Month_No
100 ORDER BY Month_No
101
```

Result Grid

Month_No	Total_Profit
1	9132.6803
2	10191.132999999993
3	28356.608900000004
4	11238.3455
5	22291.225999999995
6	21316.756000000023
7	13330.410999999978
8	21544.3955
9	35955.512900000006
10	31697.918000000003
11	34806.026799999993
12	42996.737900000004

Result 34 x

Output

Action Output

#	Time	Action	Message	Duration / Fech
3	00:16:12	SELECT Region, SUM(Sales) AS Total_Sales FROM superstore GROUP BY Region ORDER BY Total_Sales DE...	4 row(s) returned	0.015 sec / 0.000 sec
4	00:18:29	SELECT MONTHSTR_TO_DATE('Order Date', '%m/%d/%Y') AS Month_No, SUM(Sales) AS Total_Sales FRO...	12 row(s) returned	0.015 sec / 0.000 sec
5	00:20:24	SELECT MONTHSTR_TO_DATE('Order Date', '%m/%d/%Y') AS Month_No, SUM(Profit) AS Total_Profit FROM...	12 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

Q20. Duplicate Order IDs

MySQL Workbench

Local instance MySQL80

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Navigator

Assignment2solutions_Q1_Q27 SQL File 3

Limit to 1000 rows

```
101
102
103 /* Q24. Duplicate Order IDs */
104
105 SELECT
106     'Order ID',
107     COUNT(*) AS Duplicate_Count
108 FROM superstore
109 GROUP BY 'Order ID'
110 HAVING COUNT(*) > 1
111
```

Result Grid

Order ID	Duplicate_Count
CA-2016-152156	2
US-2015-108966	2
CA-2014-115812	7
US-2015-118983	2
CA-2014-143336	3
CA-2016-137330	2
CA-2016-121755	2
US-2015-150630	6
US-2016-117590	2
CA-2015-117915	4
CA-2016-118255	2
CA-2016-169194	2

Result 35 x

Output

Action Output

#	Time	Action	Message	Duration / Fech
4	00:18:29	SELECT MONTHSTR_TO_DATE('Order Date', '%m/%d/%Y') AS Month_No, SUM(Sales) AS Total_Sales FRO...	12 row(s) returned	0.015 sec / 0.000 sec
5	00:20:24	SELECT MONTHSTR_TO_DATE('Order Date', '%m/%d/%Y') AS Month_No, SUM(Profit) AS Total_Profit FROM...	12 row(s) returned	0.015 sec / 0.000 sec
6	00:22:43	SELECT 'Order ID', COUNT(*) AS Duplicate_Count FROM superstore GROUP BY 'Order ID' HAVING COUNT(*) ...	1000 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

- Filter objects
- celestial_week2
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
 - Tables
 - Views
 - Stored Procedures
 - Functions
- superstore_analysis
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sys
- world

Assignment2solutions_Q1_Q27 SQL File 17

Limit to 1000 rows

```
107 COUNT(*) AS Duplicate_Count
108 FROM superstore
109 GROUP BY "Order ID"
110 HAVING COUNT(*) > 1;
111
112 /* Q25. Check Missing Sales */
113
114 SELECT *
115 FROM superstore
116 WHERE Sales IS NULL;
```

Result Grid

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount
--------	----------	------------	-----------	-----------	-------------	---------------	---------	---------	------	-------	-------------	--------	------------	----------	--------------	--------------	-------	----------	----------

Administration Schemas

Information

Schema: superstore_analysis

superstore 36 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	00:20:24	SELECT MONTHSTR_TO_DATE(Order Date, "mm/dd/yyyy") AS Month_No, SUM(Profit) AS Total_Profit FROM...	12 row(s) returned	0.015 sec / 0.000 sec
6	00:22:43	SELECT "Order ID", COUNT(*) AS Duplicate_Count FROM superstore GROUP BY "Order ID" HAVING COUNT(*) ...	1000 row(s) returned	0.016 sec / 0.000 sec
7	00:23:41	SELECT * FROM superstore WHERE Sales IS NULL LIMIT 0, 1000	0 row(s) returned	0.016 sec / 0.000 sec

Object Info Session